Air levitation control system

Precision control of a levitating ball using an advanced air flow feedback system

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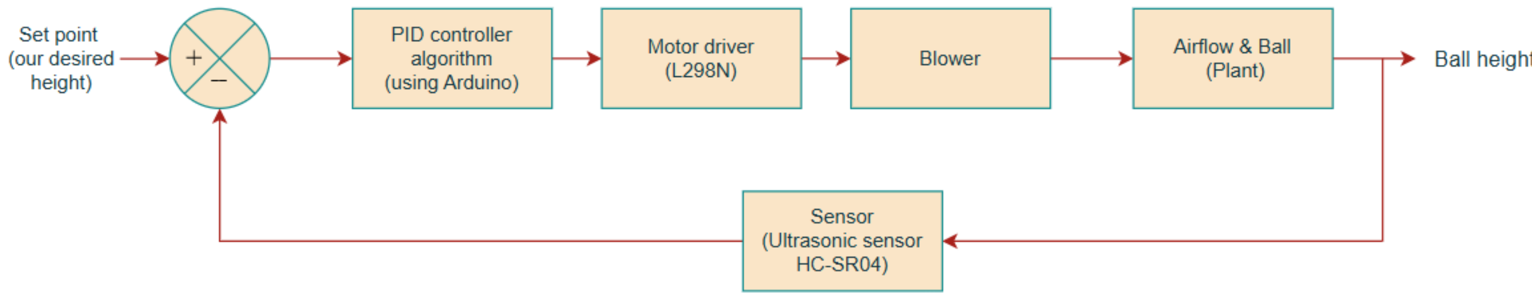
Introduction :

Air levitation (aerodynamic levitation) is the process of suspending an object in a stable position without mechanical contact by generating a controlled upward airflow that exactly counteracts gravitational force. In the ball-and-pipe control system , a lightweight ball (typically a ping-pong) is levitated inside a vertical transparent pipe. A blower at the base produces upward airflow, while the ball's vertical position is continuously measured by a non-contact distance sensor (infrared or ultrasonic). The system is inherently open-loop unstable and exhibits nonlinear dynamics due to the relationship between airflow velocity, pressure distribution (Bernoulli principle), and ball position. Feedback control adjusts the blower speed via PWM to maintain the ball at any desired height.

literature survey :

Air levitation systems are widely used in academic laboratories to demonstrate modern control techniques, including PID control, state-space modeling, and digital control. Beyond the classroom, similar systems find applications in magnetic and pneumatic levitation research, precision manufacturing and handling, and in teaching the behavior of nonlinear and inherently unstable systems. Previous studies have shown that the levitation force acting on the ball is a nonlinear function of airflow velocity, which makes the system inherently nonlinear. Consequently, linearization around an operating point is commonly employed to facilitate controller design.

System Block diagram:



1. Reference Input

The reference input defines the desired vertical position of the ball inside the pipe. It can be a fixed height for steady-state operation or a varying signal to evaluate the system's dynamic response and tracking performance.

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2. PID Controller

The PID controller, implemented on the Arduino, computes the control signal based on the error between the desired height and the actual ball position. By continuously adjusting the blower speed, it stabilizes the ball at the set point. PID control is preferred for its simplicity and effectiveness in managing nonlinear, unstable systems.

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3. Microcontroller (Arduino)

The Arduino acts as the system's central processing unit. Its main functions include: Reading real-time ball position from the ultrasonic sensor, executing the PID control algorithm and generating PWM signals to control the blower motor

Arduino is chosen for its ease of programming and compatibility with sensors and actuators.

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4. Motor Driver (L298N)

The L298N motor driver interfaces between the Arduino and the blower motor. It provides the necessary current to drive the motor while allowing precise speed control via PWM signals from the Arduino.

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5. Blower / Fan

The blower converts electrical energy into airflow. The speed of the motor determines the airflow rate, which directly influences the lifting force applied to the ball. Controlled airflow allows precise regulation of the ball height.

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6. Airflow & Ball (Plant)

The ball is levitated by the airflow inside the vertical pipe. The lifting force depends nonlinearly on airflow velocity and ball properties. The system dynamics are inherently nonlinear and unstable, making feedback control essential for stable operation.

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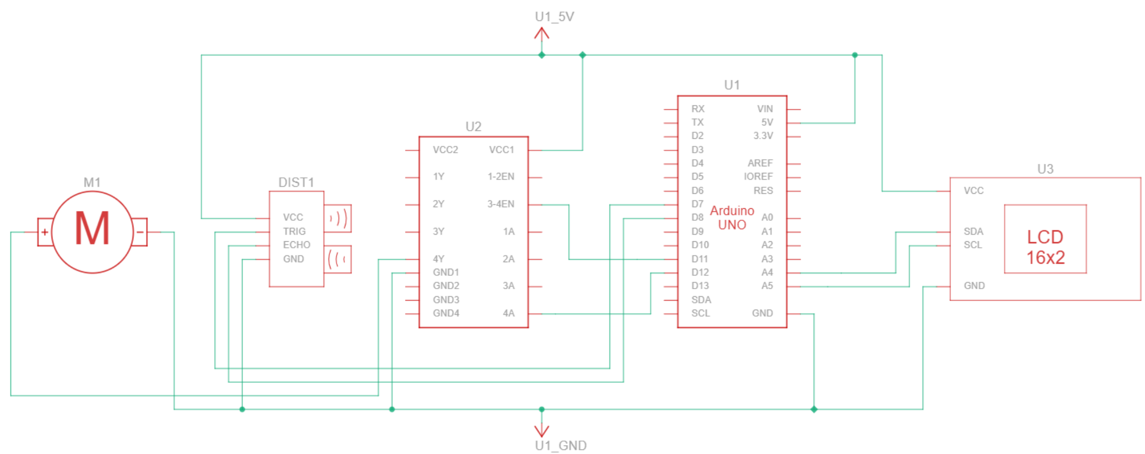
7. Distance Sensor (Ultrasonic Sensor HC-SR04)

The ultrasonic sensor measures the vertical position of the ball in real time. This feedback signal is sent to the Arduino to close the control loop, enabling accurate and stable height regulation.

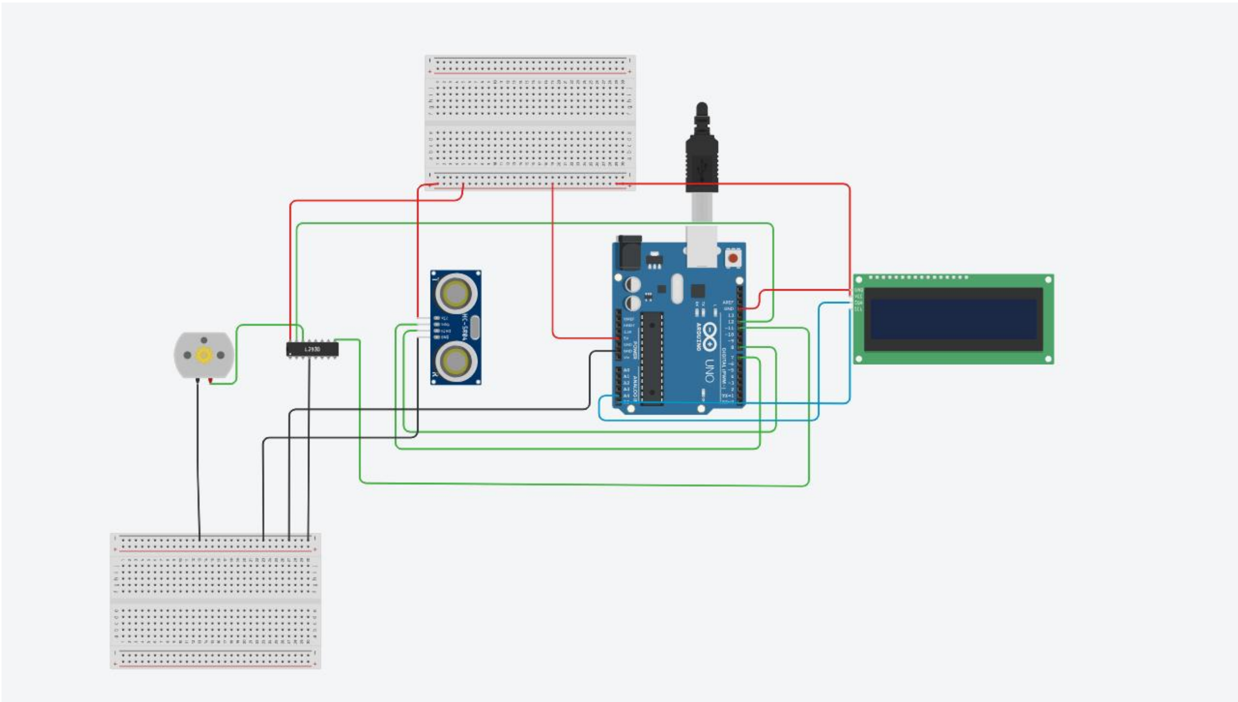
Schematic

Schematic Diagram:



-The provided schematic serves as the foundational blueprint for the system's hardware layer. It defines the closed-loop control architecture where the Arduino Uno (U1) processes distance data from the ultrasonic sensor (DIST3) to calculate the error relative to a user-defined setpoint. The resulting Pulse Width Modulation (PWM) signal is sent to the L298N driver (U2) to regulate the power delivered to the blower motor (M1), thereby modulating the upward aerodynamic force required for stable levitation.

Pictorial Wiring Diagram:



Controlling process :

Controlling a system is the perennial monitoring of the system which includes revision and correction in order to gain the control objectives and process as planned. In this project the control objective is to impel the ball to track a reference trajectory by regulating the voltage of the blower. The difference between a setpoint and the current position of the ball (position Error) is the input of the controller. The output of the controller will be the control command, sent to the blower driver to control the blower’s rotation speed. Eventually, the position of the ball will be variate according to the generated airflow. Proportional-integral-derivative (PID) controller will be adopted as a model-free linear type control strategy. The PID controller is favorite for its simple functionality which allows for straightforward operation.

System model :

The system dynamic equations are nonlinear. This is due to the nonlinear description of the air stream, which subjects to the Bernoulli's equation. This equation makes a crucial prediction about the relationship between the pressure and the velocity of a moving ideal fluid:

$$p_1 + 1/2\rho v_1^2 + \rho g y_1 = p_2 + 1/2\rho v_2^2 + \rho g y_2 \tag{1}$$

where, $1/2\rho v^2$ is kinetic energy and $\rho g y$ is gravitational potential energy, p_1 and p_2 are the static pressures of air at the cross-section, ρ is the density of the following air, y_1 and y_2 are the different distances between the ball and the bottom of the pipe, v_1 and v_2 are the mean velocities of fluid flow at the cross-section. Once a fluid moves with high velocity, it has lower pressure than the same fluid at low velocity. In the experiments, one side of the Ping-pong ball is in contact with a low pressure which creates a pressure gradient, the other side of the Ping-pong ball is in contact with a higher pressure which results in the Ping-pong ball being pushed towards the region of low pressure. Placing the ball in the tube allows to achieve higher heights but the Bernoulli principle still works. This is because the tube accumulates the air increasing the speed around the ball and allow the ball to go higher. As air is blown out from the blower, it flows at high speed and this creates a region of low pressure across the top of the pipe. The still air around the ball is at a higher pressure and pushes on the ball and causes it to stay floating.

As it is shown in Eq. (2) and Eq. (3), Newton's second law gives us the dynamic equation for the air levitation ball and pipe system, which has been studied by several previous works [13-15]. Forces acting over the levitating object are the buoyancy force (F_b), drag force (F_d) and weight force of the ball $\left(F_g\right)$. Drag force tries to prevent the levitator motion and it is a function of air pressure difference and friction force of air, and Buoyancy is an upward force exerted by a fluid that opposes the weight of a plunged object. Upwards effect of the airflow and the downwards effect of the gravity will induce an up or down motion of the ball (Fig. 3).

$$m\frac{d^2y}{dt^2} = F_b + F_d - F_g \tag{2}$$

$$F_b = \rho g V_b \tag{2}$$

$$F_d = f\left(\Delta p, F_f\right) \tag{3}$$

$$F_g = mg \tag{3}$$

Where Δp is the air pressure difference, ρ is the density of air, g is the gravitational acceleration, m is the mass of the ball, V_b is the ball's volume and F_f is the friction force caused by airflow. The drag coefficient is a term that depends on Reynold's number, which, in turn, depends on the relative velocity of the ball that moves inside the flow, and the velocity of the flow.

$$F_d\left(\rho, c_d, A, v_f, y\right) = 1/2 \cdot C_d \cdot \rho \cdot A \cdot \left(v_f - \dot{y}\right)^2 \tag{4}$$

Where, v_f is the velocity of the air inside the tube, A_b is the ball's area, C_d is the so-called drag coefficient, and y is the position of the ball in the tube. By summarizing and reforming the relations mentioned above, the system's dynamic equations can be obtained by exploiting net force between the airflow force and gravitational force as:

$$m\Delta\ddot{y} = -mg + 1/2 \cdot C_d \cdot \rho \cdot A \cdot \left(v_f - \dot{y}\right)^2 + \rho g V_b \qquad (5)$$

In this study, it is assumed that cd is constant due to the small velocity of flow. The levitating ball will be in a steady state when it does not move ($\ddot{y} = \dot{y} = 0$). The IR sensor measures the distance between the top of the pipe and the ball. In this way, the sensor provides information about the position of the ball. In the following, V_{eq} defined as airspeed at the equilibrium point $\left(v_{eq} = v_f - \dot{y}\right)$. Thus, g defines through:

$$g = \frac{Cd \cdot \rho \cdot A}{2(m - \rho V_b)} \cdot v_{eq}^2 \qquad (6)$$

Finally, the dynamic equation of the process is expressed as follows:

$$\ddot{y} = g \cdot \left(\frac{m - \rho V_b}{m}\right)\left(\left(\frac{v_f - \dot{y}}{v_{eq}}\right)^2 - 1\right) \qquad (7)$$

The system can be modeled either linear or nonlinear. Application of these two states of description depends on the type of control procedures adoption for designing a controller for the setup. Regardless of that, linearization for the system is possible around the equilibrium point by using Taylor's expansion:

$$f(x) \approx f(x_0) + f'(x_0) \cdot (x - x_0) \qquad (8)$$

By assuming $x = \frac{v_f - \dot{y}}{v_{eq}}$, and expansion of the relation (7) about the point $x = 1$, it yields to:

$$\ddot{y} = \frac{2 \cdot g}{v_{eq}} \left(\frac{m - \rho V_b}{m} \right) \left(v_f - \dot{y} - v_{eq} \right) \tag{9}$$

Determining the response of a system at an operating point is a critical step in system and controller design. The identification of the process transfer function is carried out in the frequency domain with the open loop tests performed over the system. The system is with one input and one output (SISO). The input signal is wind speed generated by blower and output is an accretion of the position of the ball. Assuming the system is well described by the linearized model, the transfer function between ball position and wind speed is:

$$\frac{y(s)}{v(s)} = \frac{1}{s} \frac{b}{(s + b)} \tag{10}$$

Where $v(s)$ and $y(s)$ are wind speed and increment of ball's position about the equilibrium point, respectively and $b = 2g(m - \rho V_b)/mveq$. Considering the fan can be modeled as a first-order process, the transfer function between the input voltage and the wind speed is represented as:

$$\frac{v(s)}{u(s)} = \frac{k_v}{\tau s + 1} \tag{11}$$

where $v(s), u(s)$, are wind speed and input voltage, respectively. In addition, k_v is the sensitivity gain that relates the input voltage to the wind speed at steady state, and τ is the time constant of the fan. It should be noted that there be existed a delay in the operator, as well as measurement delay, in the ball and pipe system, which needs to be included in the model. The fan used in this setup has electronic components that can bring out a delay in the time of performing the command. Moreover. A lowpass filter has also been used to reduce the signal noise of the measurement, which results in a delay in the system. It is necessary to mention, the values of these delays are not clear and are estimated at the identification phase. In the present paper, the effect of the two delays is assumed to be T_d, cumulatively.

$$e^{-T_d s} = \frac{1 - T_d s}{T_d s + 1} \tag{12}$$

Finally, the transfer function of the entire system defines as follows:

$$G(s) = \frac{y(s)}{u(s)} = \frac{b \cdot k_{v.}(1 - T_d s)}{s(s + b)(\tau s + 1)(T_d s + 1)} \tag{13}$$

However, this function does not describe the behavior of the system, solely, due to the limited length of the pipe, it is necessary to add a saturation block after the function. The saturation function has a lower bound of 0 and an upper bound of 40 cm . Also, the behavior of the system indicates that the ball does not move with small values of the input, so adding a dead-zone block after the input in the model will make the system performance more precise.

Components:

L298 Module Red Board (Dual H-Bridge Motor Driver Using L298N) 75.00 EGP
Categories: Robotic | Robotics Accessories DC Motor Drivers Motor Drivers & Controllers IC's

Ultrasonic Sensor HC-SR04 40.00 EGP
Distance Measurement Sensor Module
Categories: Robotic | Robotics Accessories Sensors

3D Printer 5015 Radial Turbo Blower 12Vdc Fan (SKU#1014.12V) 50.00 EGP
Categories: 3d Printer Parts Fans | Fluid Control | Pumps

Character LCD 2x16 Blue 1602 60.00 EGP
Category: Character LCD

AC-DC 5V 700mA 3.5W Step Down Isolated Switching Power Supply Module 75.00 EGP
Categories: DC/DC & AC/DC Converters Power Supply SMPS

Arduino UNO Rev3 (A65) Original Chips - Clone 450.00 EGP
Original MEGA16U2 & ATmega328 Chips Stable & Durable Performance
Category: Arduino Boards & Shields

PH60 - 20cm Male to Female 40 Jumper Wires Set 35.00 EGP
Multicolor 40-pin Dupont flat ribbon wires السعر للشريط 40 جنبر
Categories: Arduino Boards & Shields Breadboard & Accessories Cables & Crocodile Connectors

PH61 - 20cm Male to Male 40 Jumper Wires Set 35.00 EGP
Multicolor 40-pin Dupont flat ribbon wires السعر للشريط 40 جنبر
Categories: Arduino Boards & Shields Breadboard & Accessories Cables & Crocodile Connectors

BB-01 Breadboard 830 Tie Point 35.00 EGP
MB102
Category: Breadboard & Accessories

Ping Pong ball diameter = 40mm

So, preferred tube dimensions:

inner diameter=42-50 mm, length=40-50 cm